

CHONG YOU

PROFESSIONAL EXPERIENCES

GOOGLE DEEPMIND GENAI UNIT, NYC — Staff Research Scientist

Aug. 2025 — Apr. 2026

Reimagine retrieval & ranking with generative models

- **Project: In-context Ranking for Commercial Intelligence**

Developed theory of LLM capacity and algorithms for improving LLM in-context ranking capabilities

GOOGLE RESEARCH, NYC — Senior Research Scientist

Feb. 2021 — Jul. 2025

Led cross-org efforts on 1) efficiency in LLM and 2) retrieval for Ads

- **Project: Activation Sparsity for Gemini Efficiency**

Discovered *Lazy Neuron*, seminal work on LLM activation sparsity; Launched in Gemini with *Spark Transformer*; **Google Perfy Award**

- **Project: Hierarchical Retrieval for Ads**

Formulated product requirement as *Hierarchical Retrieval*, developed theory and algorithm for improved quality

- **Project: Retrieval Models for Google Search, Search Ads & Play Search**

Delivered four retrieval models for Search, Search Ads, and Play Search; **Google Research Tech Impact Award**

UNIVERSITY OF CALIFORNIA, BERKELEY — Postdoc Scholar

Oct. 2018 — Jan. 2021

Led multiple efforts on principled deep learning

- **Project: Towards “Whitebox” Network Architectures**

Pioneered interpretable deep network architectures from first principles, e.g., *ReduNet*, *IsoNet*, *DELVE*

- **Project: Understanding Generalization of Deep Models via Bias-Variance Analysis**

Developed *Unimodal variance principle* that demystified “double-descent” and deep model generalization

- **Project: Improving Robustness of Deep Models via Implicit Bias**

Initiated application of implicit bias to image denoising and label denoising achieving SOTA performance

- **Project: Understanding Deep Embeddings through Optimization Landscape Analysis**

Propelled a sequence of works on the mathematical foundations of deep features

JOHNS HOPKINS UNIVERSITY, BALTIMORE — Research Assistant

Sep. 2012 — Oct. 2018

- **Project: Theory and Practice of Multi-subspace Learning**

Developed theoretical foundations for multi-subspace learning and practical algorithms for large-scale, corrupted, and imbalanced data.

EDUCATION

Johns Hopkins University, Baltimore

- Ph.D. in Electrical & Computer Engineering

Sep. 2012 — Oct. 2018

Dissertation: *Sparse methods for learning multiple subspaces from large-scale, corrupted and imbalanced data*

Committee: René Vidal (advisor), Daniel P. Robinson, Trac Tran, Vishal Patel, Yi Ma

Peking University, Beijing

- M.S. in Electrical Engineering

Sep. 2009 — Jul. 2012

- B.S. in Electrical Engineering

Sep. 2005 — Jul. 2009

- B.S. in Applied Mathematics

Sep. 2006 — Jul. 2009

HONORS & AWARDS

Google Perfy Award, Gold (the highest) Tier	2025
Awarded to our work on activation sparsity for “addressing high-value efficiency issues”	
Google Tech Impact Award	2022
Awarded to our work on Google Ads for “incredible impact, tech excellence and strong collaborations”	
Reviewer Awards for CVPR, ICML, NeurIPS and ICLR	2018 - 2022
Recognized as Outstanding Reviewer for <i>CVPR 2019</i> , <i>CVPR 2020</i> , <i>ICLR 2021</i> , and <i>ICML 2022</i> ; Top 5% Reviewer for <i>ICML 2019</i> ; Top 200 Reviewer for <i>NeurIPS 2018</i>	
MINDS Doctoral Dissertation Award	2019
Awarded annually to one dissertation in data science at the Johns Hopkins University	
Editor’s Choice Award	2019
Awarded to our paper published at Demographic Research	

PUBLICATIONS (SELECTED BY THEME)

Activation Sparsity for LLM Efficiency

Established activation sparsity as a production-ready paradigm for LLM efficiency, deployed in Gemini / TPU

- [1] Spark Transformer: Reactivating Sparsity in FFN and Attention — NeurIPS’25
- [2] HiRE: High Recall Approximate Top- k Estimation for Efficient LLM Inference — ’24
- [3] The Lazy Neuron Phenomenon: On Emergence of Activation Sparsity in Transformers — ICLR’23

Information Retrieval via Long-context LLMs

Redefined information retrieval in the era of LLMs and tackled fundamental road blockers

- [1] Scalable In-context Ranking with Generative Models — NeurIPS’25
- [2] Hierarchical Retrieval: The Geometry and a Pretrain-Finetune Recipe — NeurIPS’25
- [3] Efficient and Asymptotically Unbiased Constrained Decoding for Large Language Models — AISTATS’25
- [4] Functional Interpolation for Relative Positions Improves Long Context Transformers — ICLR’24

Transparent Architectures

Shaped modern understanding of deep representation geometry and working mechanisms of deep neural networks

- [1] Generalized Neural Collapse for a Large Number of Classes — ICML’24
- [2] Deep Self-expressive Learning — CPAL’24
- [3] Revisiting Sparse Convolutional Model for Visual Recognition — NeurIPS’22
- [4] Are All Losses Created Equal: A Neural Collapse Perspective — NeurIPS’22
- [5] ReduNet: A White-box Deep Network from the Principle of Maximizing Rate Reduction — JMLR’22
- [6] A Geometric Analysis of Neural Collapse with Unconstrained Features — NeurIPS’21

Principles of Generalization, Optimization, and Robustness for Deep Models

Demystified the success of deep models and improved robustness by discovering new mathematical principles

- [1] It’s an Alignment, Not a Trade-off: Revisiting Bias and Variance in Deep Models — ICLR’24
- [2] Robust Training under Label Noise by Over-parameterization — ICML’22
- [3] Robust Learning via Implicit Bias of Discrepant Learning Rates for Double Over-parameterization — NeurIPS’20
- [4] Deep Isometric Learning for Visual Recognition — ICML’20
- [5] Rethinking Bias-Variance Trade-off for Generalization of Neural Networks — ICML’20

PUBLICATIONS (FULL LIST)

Fields: Machine learning (e.g., *ICLR*, *NeurIPS*, *ICML*, *TPAMI*), computer vision (e.g., *CVPR*, *ICCV*, *ECCV*), signal processing (e.g., *TIP*, *JSTSP*), optimization and demography.

(α - β indicates alphabetical authorship order; * indicates equal contribution; † indicates equal advising)

Preprints and Technical Reports

- [1] Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, LongContext, and Next Generation Agentic Capabilities.
Gemini Team, 2025
- [2] Autoregressive Ranking: Bridging the Gap Between Dual and Cross Encoders
Benjamin Rozonoyer, Chong You, Michael Boratko, Himanshu Jain, Nilesch Gupta, Srinadh Bhojanapalli, Andrew McCallum, Felix Yu, 2026
- [3] HiRE: High Recall Approximate Top- k Estimation for Efficient LLM Inference.
Yashas Samaga B L, Varun Yerram, Chong You, Srinadh Bhojanapalli, Sanjiv Kumar, Prateek Jain, Praneeth Netrapalli, 2024
- [4] Basis Pursuit and Orthogonal Matching Pursuit for Subspace-preserving Recovery: Theoretical Analysis.
($\alpha - \beta$) Daniel P. Robinson, René Vidal, Chong You, 2020

Published Conference and Workshop Articles

- [1] **[NeurIPS'25]** Spark Transformer: Reactivating Sparsity in FFN and Attention.
Chong You*, Kan Wu*, Zhipeng Jia*, Lin Chen*, Srinadh Bhojanapalli, Jiaxian Guo, Utku Evci, Jan Wassenberg, Praneeth Netrapalli, Jeremiah J. Willcock, Suvinay Subramanian, Felix Chern, Alek Andreev, Shreya Pathak, Felix Yu, Prateek Jain, David E. Culler, Henry M. Levy, Sanjiv Kumar.
Neural Information Processing Systems
- [2] **[NeurIPS'25]** Hierarchical Retrieval: The Geometry and a Pretrain-Finetune Recipe.
Chong You, Rajesh Jayaram, Ananda Theertha Suresh, Robin Nittka, Felix X. Yu, Sanjiv Kumar.
Neural Information Processing Systems
- [3] **[NeurIPS'25]** Scalable In-context Ranking with Generative Models.
Nilesch Gupta, Chong You, Srinadh Bhojanapalli, Sanjiv Kumar, Inderjit S Dhillon, Felix Yu.
Neural Information Processing Systems
- [4] **[AISTATS'25]** Efficient and Asymptotically Unbiased Constrained Decoding for Large Language Models.
Haotian Ye, Himanshu Jain, Chong You, Ananda Theertha Suresh, Haowei Lin, James Zou, Felix Yu.
International Conference on Artificial Intelligence and Statistics
- [5] **[ICML'24]** Generalized Neural Collapse for a Large Number of Classes.
Jiachen Jiang*, Jinxin Zhou*, Peng Wang, Qing Qu, Dustin Mixon, Chong You†, Zhihui Zhu†.
International Conference on Machine Learning
- [6] **[ICLR'24]** It's an Alignment, Not a Trade-off: Revisiting Bias and Variance in Deep Models.
Lin Chen, Michal Lukasik, Wittawat Jitkrittum, Chong You, Sanjiv Kumar.
International Conference on Learning Representations
- [7] **[ICLR'24]** Functional Interpolation for Relative Positions Improves Long Context Transformers.
Shanda Li, Chong You, Guru Guruganesh, Joshua Ainslie, Santiago Ontanon, Manzil Zaheer, Sumit Sanghai, Yiming Yang, Sanjiv Kumar, Srinadh Bhojanapalli.
International Conference on Learning Representations
- [8] **[CPAL'24]** Deep Self-expressive Learning.
Chen Zhao, Chun-Guang Li, Wei He, Chong You.
Conference on Parsimony and Learning

- [9] **[ICLR'23]** Teacher Guided Training: An Efficient Framework for Knowledge Transfer.
Manzil Zaheer, Ankit Singh Rawat, Seungyeon Kim, Chong You, Himanshu Jain, Andreas Veit, Rob Fergus, Sanjiv Kumar.
International Conference on Learning Representations
- [10] **[ICLR'23]** The Lazy Neuron Phenomenon: On Emergence of Activation Sparsity in Transformers.
Zonglin Li*, Chong You*, Srinadh Bhojanapalli, Daliang Li, Ankit Singh Rawat, Sashank J. Reddi, Ke Ye, Felix Chern, Felix Yu, Ruiqi Guo, Sanjiv Kumar.
International Conference on Learning Representations
- [11] **[NeurIPS'22]** Revisiting Sparse Convolutional Model for Visual Recognition.
Xili Dai, Mingyang Li, Pengyuan Zhai, Shengbang Tong, Xingjian Gao, Shao-Lun Huang, Zhihui Zhu, Chong You, Yi Ma.
Neural Information Processing Systems
- [12] **[NeurIPS'22]** Are All Losses Created Equal: A Neural Collapse Perspective.
Jinxin Zhou, Chong You, Xiao Li, Kangning Liu, Sheng Liu, Qing Qu, Zhihui Zhu.
Neural Information Processing Systems
- [13] **[ICML'22]** Robust Training under Label Noise by Over-parameterization.
Sheng Liu, Zhihui Zhu, Qing Qu, Chong You.
International Conference on Machine Learning
- [14] **[ICML'22]** On the Optimization Landscape of Neural Collapse under MSE Loss: Global Optimality with Unconstrained Features.
Jinxin Zhou, Xiao Li, Tianyu Ding, Chong You, Qing Qu, Zhihui Zhu.
International Conference on Machine Learning
- [15] **[NeurIPS'21]** A Geometric Analysis of Neural Collapse with Unconstrained Features.
Zhihui Zhu*, Tianyu Ding*, Jinxin Zhou, Xiao Li, Chong You, Jeremias Sulam, Qing Qu.
Neural Information Processing Systems **[Spotlight presentation with acceptance rate 3%]**
- [16] **[NeurIPS'21]** Convolutional Normalization: Improving Deep Convolutional Network Robustness and Training.
Sheng Liu*, Xiao Li*, Yuexiang Zhai, Chong You, Zhihui Zhu, Carlos Fernandez-Granda, Qing Qu.
Neural Information Processing Systems
- [17] **[ICML'21]** A Nullspace Property for Subspace-Preserving Recovery.
Mustafa Devrim Kaba, Chong You, Daniel P. Robinson, Enrique Mallada, and René Vidal.
International Conference on Machine Learning
- [18] **[CVPR'21]** Learning a Self-Expressive Network for Subspace Clustering.
Shangzhi Zhang, Chong You, René Vidal, Chun-Guang Li.
IEEE Conference on Computer Vision and Pattern Recognition
- [19] **[CVPR'21]** Incremental Learning via Rate Reduction.
Ziyang Wu*, Christina Baek*, Chong You, Yi Ma.
IEEE Conference on Computer Vision and Pattern Recognition, also at
ICML 2021 Workshop on Theory and Foundation of Continual Learning
- [20] **[ICLR'21]** A Critique of Self-expressive Deep Subspace Clustering.
Benjamin Haeffele, Chong You, René Vidal.
International Conference on Learning Representations
- [21] **[NeurIPS'20 Workshop]** Deep Networks from the Principle of Rate Reduction.
Ryan Chan*, Yaodong Yu*, Chong You*, Haozhi Qi, John Wright, Yi Ma.
Beyond Backprop Workshop at Neural Information Processing Systems
- [22] **[NeurIPS'20]** Robust Learning via Implicit Bias of Discrepant Learning Rates for Double Over-parameterization.
Chong You*, Zhihui Zhu*, Qing Qu, Yi Ma.
Neural Information Processing Systems **[Spotlight presentation with acceptance rate 4%]**

- [23] **[NeurIPS'20]** Learning Diverse and Discriminative Representations via the Principle of Maximal Coding Rate Reduction.
Yaodong Yu*, Ryan Chan*, Chong You, Chaobing Song, Yi Ma.
Neural Information Processing Systems
- [24] **[ICML'20]** Rethinking Bias-Variance Trade-off for Generalization of Neural Networks.
Zitong Yang*, Yaodong Yu*, Chong You, Jacob Steinhardt, Yi Ma.
International Conference on Machine Learning
- [25] **[ICML'20]** Deep Isometric Learning for Visual Recognition.
Haozhi Qi, Chong You, Xiaolong Wang, Yi Ma, Jitendra Malik.
International Conference on Machine Learning
- [26] **[CVPR'20]** Stochastic Sparse Subspace Clustering.
Ying Chen, Chun-Guang Li, Chong You.
IEEE Conference on Computer Vision and Pattern Recognition
- [27] **[ICCV'19]** Is an Affine Constraint Needed for Affine Subspace Clustering?
Chong You, Chun-Guang Li, Daniel P. Robinson, René Vidal.
International Conference on Computer Vision
- [28] **[ICCV'19 Workshop]** Classifying and Comparing Approaches to Subspace Clustering with Missing Data.
Connor Lane, Ron Boger, Chong You, Manolis Tsakiris, Benjamin Haeffele, René Vidal.
International Workshop on Robust Subspace Learning and Applications in Computer Vision, in conjunction with International Conference on Computer Vision
- [29] **[CVPR'19]** Self-Supervised Convolutional Subspace Clustering Network.
Junjian Zhang, Chun-Guang Li, Chong You, Xianbiao Qi, Honggang Zhang, Jun Guo, Zhouchen Lin.
IEEE Conference on Computer Vision and Pattern Recognition
- [30] **[ECCV'18]** A Scalable Exemplar-based Subspace Clustering Algorithm for Class-Imbalanced Data.
Chong You, Chi Li, Daniel P. Robinson, René Vidal.
European Conference on Computer Vision
- [31] **[CVPR'17]** Provable Self-Representation Based Outlier Detection in a Union of Subspaces.
Chong You, Daniel P. Robinson, René Vidal.
IEEE Conference on Computer Vision and Pattern Recognition
[Spotlight presentation with acceptance rate 8%]
- [32] **[CVPR'16]** Scalable Sparse Subspace Clustering by Orthogonal Matching Pursuit.
Chong You, Daniel P. Robinson, René Vidal.
IEEE Conference on Computer Vision and Pattern Recognition
[Oral presentation with acceptance rate 3.9%]
- [33] **[CVPR'16]** Oracle Based Active Set Algorithm for Scalable Elastic Net Subspace Clustering.
Chong You, Chun-Guang Li, Daniel P. Robinson, René Vidal.
IEEE Conference on Computer Vision and Pattern Recognition
[Oral presentation with acceptance rate 3.9%]
- [34] **[ICML'15]** Geometric Conditions for Subspace-Sparse Recovery.
Chong You, René Vidal.
International Conference on Machine Learning
- [35] **[SPARS'15]** On Sufficient Conditions for Affine Sparse Subspace Clustering.
Chun-Guang Li, Chong You, René Vidal.
Signal Processing with Adaptive Sparse Structured Representations
- [36] **[SPARS'15]** Geometric Conditions for Subspace-Sparse Recovery.
Chong You, René Vidal.
Signal Processing with Adaptive Sparse Structured Representations.

Published Journal Articles

- [1] **[JMLR]** Recovery and Generalization in Over-Realized Dictionary Learning.
Jeremias Sulam, Chong You, Zhihui Zhu.
Journal of Machine Learning Research, 2022
- [2] **[JMLR]** ReduNet: A White-box Deep Network from the Principle of Maximizing Rate Reduction.
Ryan Chan*, Yaodong Yu*, Chong You*, Haozhi Qi, John Wright, Yi Ma.
Journal of Machine Learning Research, 2022
- [3] **[TPAMI]** Self-Representation Based Unsupervised Exemplar Selection in a Union of Subspaces.
Chong You, Chi Li, Daniel P. Robinson, René Vidal.
IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022
- [4] Improving Age Measurement in Low- and Middle-income Countries through Computer Vision: A Test in Senegal.
Stéphane Helleringer, Chong You, Laurence Fleury, Laetitia Douillot, Cheikh Tidiane Ndiaye, Valerie Delaunay, René Vidal.
Demographic Research, 2019 **[Editor's choice]**
- [5] **[J-STSP]** On Geometric Analysis of Affine Sparse Subspace Clustering.
Chun-Guang Li*, Chong You*, René Vidal.
IEEE Journal of Selected Topics in Signal Processing, 2018
- [6] A Nonconvex Formulation for Low Rank Subspace Clustering: Algorithms and Convergence Analysis.
Hao Jiang*, Daniel P. Robinson*, René Vidal*, Chong You*.
Computational Optimization and Applications, 2018
- [7] **[TIP]** Structured Sparse Subspace Clustering: A Joint Affinity Learning and Subspace Clustering Framework.
Chun-Guang Li, Chong You, René Vidal.
IEEE Transactions on Image Processing, 2017

Invited Publications

- [1] A Divide-and-Conquer Framework for Large-Scale Subspace Clustering.
Chong You, Claire Donnat, Daniel P. Robinson, René Vidal.
Asilomar Conference on Signals, Systems, and Computers, 2016

TALKS & PRESENTATIONS

How Many FLOPS is a Token Worth?

- *Zhiyuan Community [Remote]* *Jul. 2025*
- *Workshop on Mathematics of Deep Learning, Oaxaca, Mexico* *Jun. 2024*

Large Models are Parsimonious Learners?

- *Google Workshop on Sparsity and Adaptive Computation [Online]* *Oct. 2022*

Deep Self-expressive Model

- *SIAM Conference on Imaging Science [Remote]* *Mar. 2022*

Robust Recovery via Double Over-parameterization

- *SIAM Conference on Mathematics of Data Science [Remote]* *Sep. 2022*
- *Communications and Signal Processing Seminar at University of Michigan, MI, USA* *Sep. 2022*
- *International Conference on Continuous Optimization, PA, USA* *Jul. 2022*
- *Advances in Neural Information Processing Systems (NeurIPS) [Remote]* *Dec. 2020*

Low-dimensional Modeling for Data-driven Learning

· Lawrence Livermore National Laboratory [Remote]	Nov. 2020
Learning a Union of Subspaces from Big and Corrupted Data	
· SIAM Conference on Mathematics of Data Science [Remote]	Jun. 2020
· 2019 MINDS Symposium on the Foundations of Data Science, MD, USA	Nov. 2019
· International Conference on Pattern Recognition (ICPR), Beijing, China [Remote]	Aug. 2018
· Center on Frontiers of Computing Studies (CFCS), Peking University, Beijing, China	Apr. 2018
· Google Inc., CA, USA	Jun. 2017
Self-supervised Convolutional Subspace Clustering Network	
· IEEE Conference on Computer Vision and Pattern Recognition (CVPR), CA, USA [Poster]	Jul. 2019
Provable Self-representation Based Outlier Detection in a Union of Subspaces	
· IEEE Conference on Computer Vision and Pattern Recognition (CVPR), HI, USA	Jul. 2017
· Mid-Atlantic Computer Vision (MACV) Workshop, PA, USA	May. 2017
Large Scale Subspace Clustering Algorithms	
· Asilomar Conference on Signals, Systems, and Computers, CA, USA	Nov. 2016
Scalable Elastic-net Subspace Clustering	
· Amazon Graduate Research Symposium, WA, USA [Poster]	Jan. 2017
· IEEE Conference on Computer Vision and Pattern Recognition (CVPR), NV, USA	Jul. 2016
· Mid-Atlantic Computer Vision (MACV) Workshop, MD, USA	May. 2016
Geometric Conditions for Subspace Sparse Recovery	
· Mid-Atlantic Computer Vision (MACV) Workshop, MD, USA [Poster]	May. 2016
· Signal Processing with Adaptive Sparse Structured Representations (SPARS) Workshop, Cambridge, UK [Poster]	Jul. 2015

ACADEMIC SERVICES

Event Organizer

Co-founder of SlowDNN Workshop (2020 - 2023) → CPAL Conference (2024 - 2025) series, the leading venue on low-dimensionality in deep models

· 2nd Conference on Parsimony and Learning, Stanford, CA	Mar. '25
· Conference on Parsimony and Learning, Hong Kong	Jan. '24
· Mini-Symposium on <i>Deep Learning with Low-Dimensional Models</i> , at SIAM Conference on Mathematics of Data Science, San Diego, CA	Sep. '22
· The 3rd Workshop on Seeking Low-dimensionality in Deep Neural Networks (SLowDNN)	Jan. '23
· The 2nd Workshop on Seeking Low-dimensionality in Deep Neural Networks (SLowDNN)	Nov. '21
· Workshop on Seeking Low-dimensionality in Deep Neural Networks (SLowDNN)	Nov. '20
· Tutorial on Subspace Clustering: Recent Advances in Algorithms, Theories and Applications, at International Conference on Pattern Recognition (ICPR), Beijing	Aug. 2018

Conference Area Chair

· Neural Information Processing Systems (NeurIPS)	'26
· International Conference on Learning Representations (ICLR)	'25, '26
· International Conference on Machine Learning (ICML)	'25, '26

Conference Reviewer (# articles reviewed)

· IEEE Conference on Computer Vision and Pattern Recognition (CVPR)	(46)	'16 - '22
· Neural Information Processing Systems (NeurIPS)	(27)	'15, '18 - '22

· International Conference on Machine Learning (ICML)	(8)	'19 - '20, '22
· European Conference on Computer Vision (ECCV)	(8)	'18
· International Conference on Computer Vision (ICCV)	(6)	'17
· British Machine Vision Conference (BMVC)	(3)	'17
· International Conference on Pattern Recognition (ICPR)	(4)	'18, '22
· International Conference on Learning Representations (ICLR)	(7)	'20 - '22

Journal Reviewer (# articles reviewed)

· IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	(12)
· IEEE Transactions on Image Processing (TIP)	(12)
· IEEE Transactions on Signal Processing (TSP)	(1)
· IEEE Transactions on Neural Networks and Learning Systems (TNNLS)	(1)
· IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)	(1)
· IEEE Transactions on Knowledge and Data Engineering (TKDE)	(1)
· IEEE Transactions on Circuits and Systems (TCAS) II: Express Briefs	(1)
· IEEE Transactions on Knowledge Discovery from Data (TKDD)	(1)
· IEEE Access	(2)
· IEEE Journal of Selected Topics in Signal Processing (J-STSP)	(5)
· SIAM Journal on Imaging Sciences (SIIMS)	(1)
· Journal of Machine Learning Research (JMLR)	(2)
· Journal of Statistical Computation and Simulation (JSCS)	(1)
· Computer Science Review	(1)
· Pattern Recognition	(6)
· Neurocomputing	(7)

TEACHING EXPERIENCES

University of California, Berkeley

· Teaching Assistant for <i>High-dimensional Data Analysis with Low-dimensional Models</i>	<i>Fall, 2019</i>
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Johns Hopkins University

· Teaching Assistant for <i>Advanced Topics in Machine Learning</i>	<i>Spring, 2016</i>
· Course Assistant for <i>Computer Vision</i>	<i>Fall, 2014</i>

Peking University

· Teaching Assistant for <i>Introduction to Satellite Navigation Systems</i>	<i>Fall, 2010</i>
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